ZapGPT-Morph: from a sentence to a verified, printable robot

No reinforcement learning: body and gait are found together by one evolutionary search

1 Prompt

"a four-legged robot that walks sideways to the right"

free English text

2 LLM compiler

local LLM, JSON schema
only allowed fields & ops

maximise -10 * dy
subject to n_legs == 4

invalid -> errors sent back
-> repair

3 One genome, 79 genes

24 body genes
torso size, # leg pairs,
segments, lengths, joint axes

55 gait genes (CPG)
q*(t) = b + A sin(2 pi f t + phi)
per joint: A, phi, b; shared f

4 Evolve body + gait

genome -> URDF -> MuJoCo, 6 s
servo: 3.8 N m, 5.4 rad/s

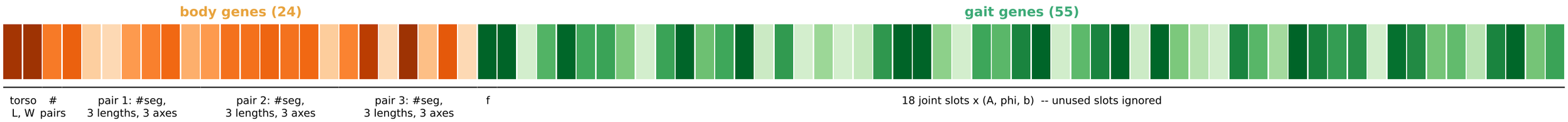
fitness = objective
- body-constraint penalty
- fall penalty (tilt > 34 deg)
worst of friction {1.0, 0.7}
GA 64 x 150 generations

5 Verify

human-written pass rules,
fixed before the run,
independent of the LLM

held-out physics:
friction 0.85, dt 1 ms

-> GIF + printable URDF



Evolved genome of the right-walking robot (colour = gene value in [0, 1]). Discrete genes (# pairs, # segments, axis) are read by binning [0, 1].

A

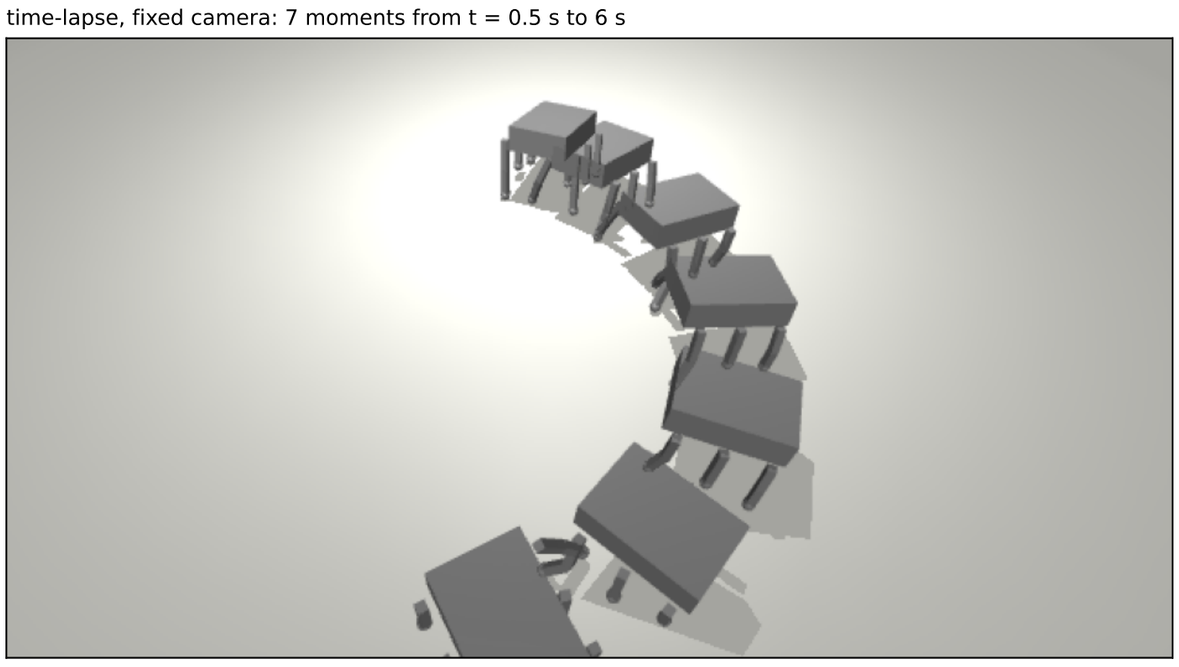
"a six-legged robot that walks forward"

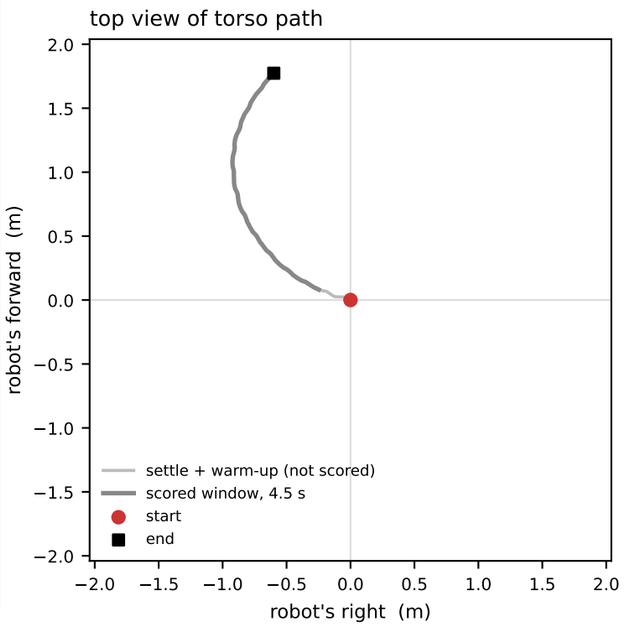
LLM spec
maximise +10*dx
subject to n_legs == 6

Evolved body
6 legs, 10 servos, segments/pair [2, 1, 2]
joint axes: 3x sideways swing, 1x fore-aft swing, 1x yaw
torso 0.33 x 0.22 m, 1.95 kg, stands 0.23 m

Behaviour (scored 4.5 s)
forward +1.70 m, right -0.35 m, turn -114 deg, no fall

Verification
body rule PASS behaviour rule PASS
held-out: friction 0.85 -> 1.64 m, dt 1 ms -> 1.64 m forward





B

"a four-legged robot that walks sideways to the right"

LLM spec
maximise -10*dy
subject to n_legs == 4

Evolved body
4 legs, 8 servos, segments/pair [2, 2]
joint axes: 4x sideways swing
torso 0.33 x 0.24 m, 1.77 kg, stands 0.19 m

Behaviour (scored 4.5 s)
forward -0.20 m, right +2.02 m, turn -9 deg, no fall

Verification
body rule PASS behaviour rule PASS
held-out: friction 0.85 -> 2.05 m, dt 1 ms -> 1.32 m right

